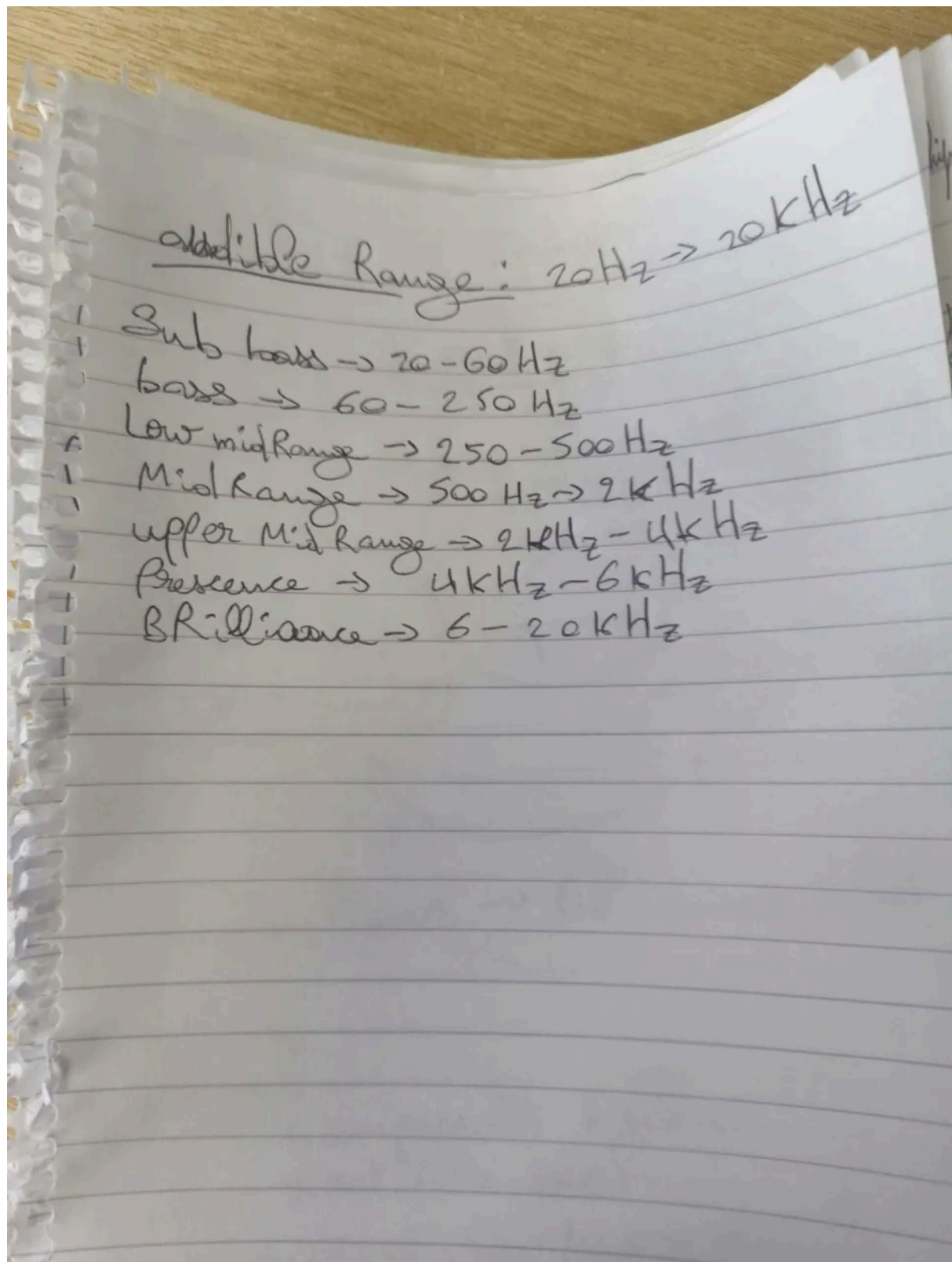


I am working on making a video about a circuit that could generate a tone so that I can then build everything from there, I want this to be a sawtooth generator but it needs to produce set frequencies corresponding to the 12 fundamental notes.

Music Theory:

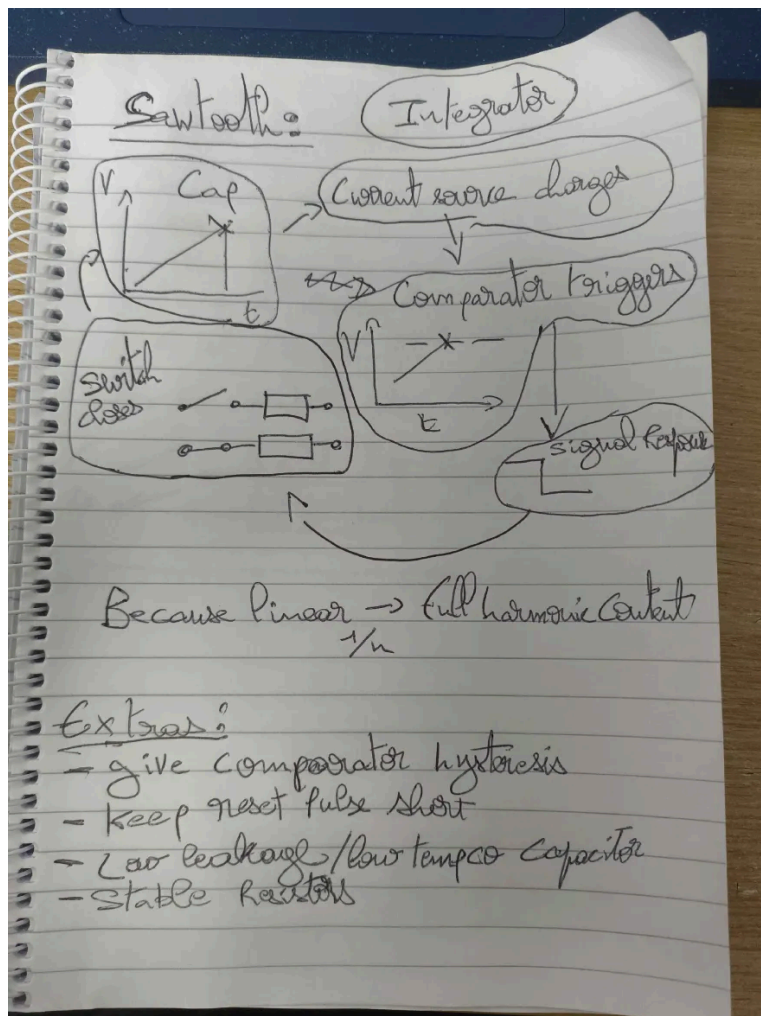
55.0 Hz	A		
61.7 Hz	B		58.3 Hz
65.4 Hz	C2		
73.4 Hz	D		69.3 Hz
82.4 Hz	E		77.8 Hz
87.3 Hz	F		
98.0 Hz	G		92.5 Hz
110.0 Hz	A		103.8 Hz
123.5 Hz	B		116.5 Hz
130.8 Hz	C3		
146.8 Hz	D		138.6 Hz
164.8 Hz	E		155.6 Hz
174.6 Hz	F		
196.0 Hz	G		185.0 Hz
220.0 Hz	A		207.7 Hz
246.9 Hz	B		233.1 Hz
261.6 Hz	C4		
293.7 Hz	D		277.2 Hz
329.6 Hz	E		311.1 Hz
349.2 Hz	F		
392.0 Hz	G		370.0 Hz

Ref1



This sawtooth generator thus has to be variable easily so that I can get it to play any note,

Sawtooth Generator:

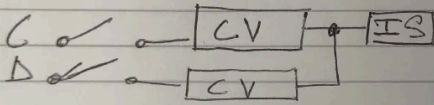


I found out that sawtooth generators usually work by having a current source charge up a capacitor, followed by a comparator triggering once the capacitor gets to a certain point. The comparator's signal then makes a switch which is responsible for discharging the capacitor activate. This discharges the capacitor so that the cycle can start again, and the frequency is then set by how quickly you charge and discharge the capacitor.

Current source:

Can be an IC, transistor current source mirror,
the $\frac{dV}{dt} = I/C$ so discharge/charge time is directly proportional to I if C is constant.

So to match the notes to current



I then looked into how current sources work for this kind of utilisation, they can be coming from a transistor current mirror circuit,

I then drew that I would need a switch per note which would lead to a certain current source being provided to the capacitor for charging,

Or if using a voltage controlled oscillator then it would be a certain voltage per switch.

how CV is generated and tuned?

- Understand what OSC used:

VCO $\rightarrow$ Voltage Controlled oscillator
PLL $\rightarrow$ Phase Locked Loop

PLL $\rightarrow$ Phase Locked loop

NESS6 used to be good $\rightarrow$ obsolete

555 $\rightarrow$ can be a good basis

L M358 $\rightarrow$ doesn't handle high freq

ICL8038/xR206 → fluorescent ICs

issue is that $\text{out} \propto \text{Input modulation}$

Whereas the notes freq relationship is exponential

So need a

need a
Logarithmic amp to provide CV

But some ICs can ~~be~~ be
are much better VCOs

MAX 2624 Good but expensive

SSS: audible spectrum: $20 \rightarrow 20 \text{ kHz}$
SSS $\rightarrow$ $1 \text{ mHz} \rightarrow 100 \text{ kHz}$

SSS $\rightarrow$ 1 mHz $\rightarrow$ 100 kHz

I ended up choosing the 555 timer for its low cost, high availability, high amount of documentation and the fact that it can handle 1mHz up to 100kHz, I learned about it from Ref3 and investigated the components listed in the article.

③

Part 2: 555 deep dive.

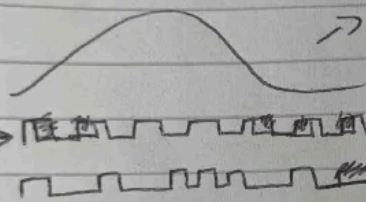
How to control output frequency:

- Pulse Width Modulation (PWM)

Section of datasheet

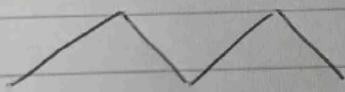
"Applying ext V or I to Cont modifies the internal ~~threshold~~ threshold and trigger Voltage".

Varies Pulse Width!!



→ if we input this in C/V
→ and a constant clock
→ the freq varies

- PPM pulse position modulator:



→ input control V

→ out freq

Varies Pulse position

So use PPM as Reference in datasheet

Monostable Mode → only one stable state, so oscillate in one state until an external control hits

I then deep dived into the 555 timer through its datasheet, I came to learn that my desired configuration would be the PPM setup, and that the frequency which is related to the positioning of the pulses could be varied by applying a control voltage to the CONT pin,

This sounded very simple and so I decided to stick with this idea and just use trimmers on the input so that I could tune the instrument by having a frequency meter on the output and tuning each note individually,

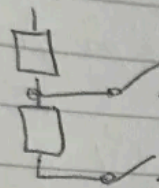
I also discovered that the opamp seems to output a 5V pk to pk signal from the graphs, this is to be tested and I believe it is dependent on the power given to the ic, so I should check the datasheet further, but I will be more than fine with my virtual ground setup because it will be ac coupled which will get rid of the dc bias and thus the signal should fit in the virtual gnd's realm.

② Part 3 - the CV & out f relationship

- it's voltage level
- how to generate CV

Button input:

Vcc



→ Amp needed in unity gain to buffer signal

CONT

Resistor ladder, use trimmers so can tune it

Relationship:

Can dig formula but easy is to get a frequency measure and tune it
Later dive into internals if want,
Separate Page !!

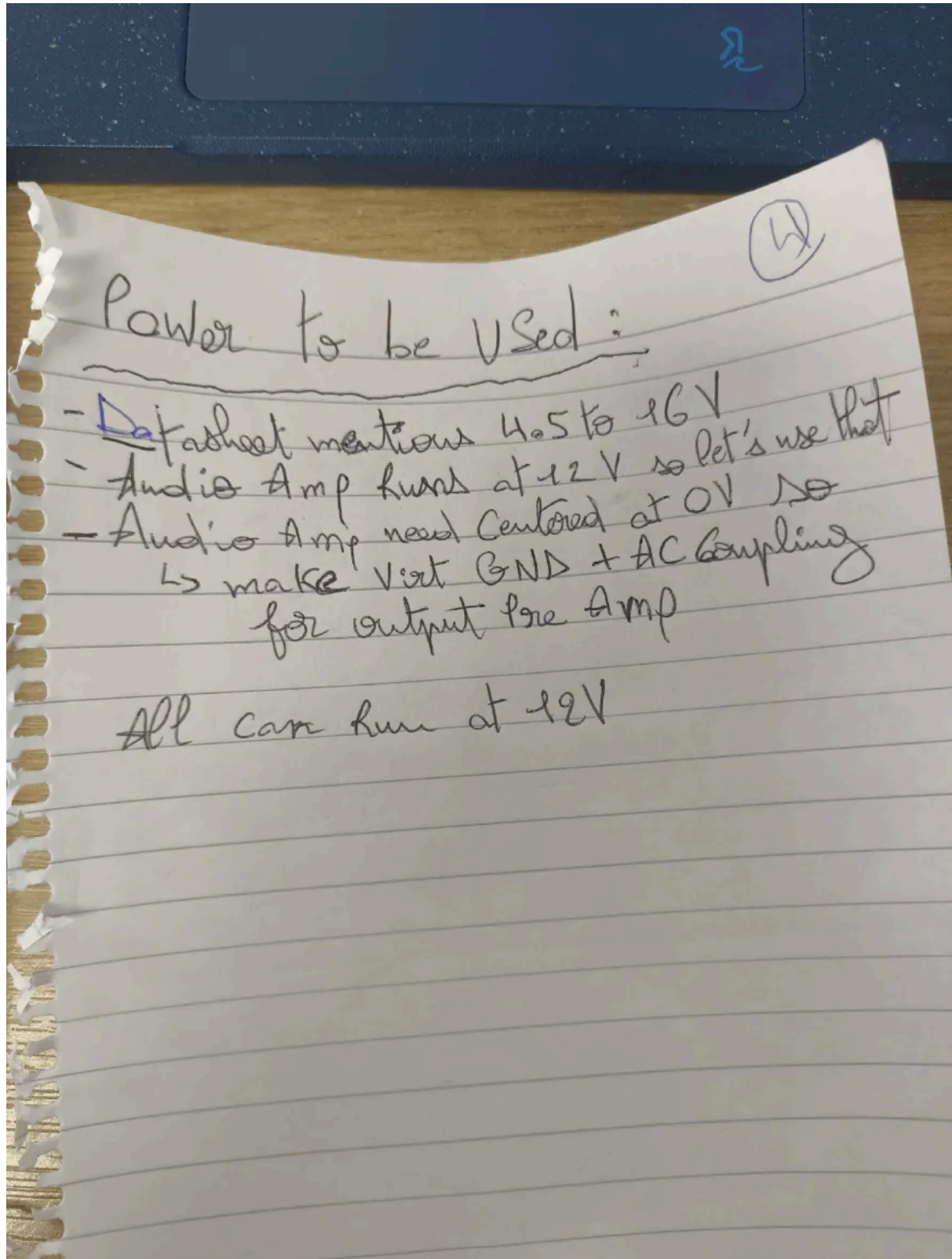
Output:

from datasheet this seems to be about 5V but should be tested
See more in part 5

Here I looked deeper into what method I would use to control the input of the 555 timer to control it's frequency,

I then thought I could analyse the formula in a deep dive format but that would have to wait as it is a complicated topic,

And lastly setup the plan regarding how I would morph the oscillator's output to fit the audio amplifier,

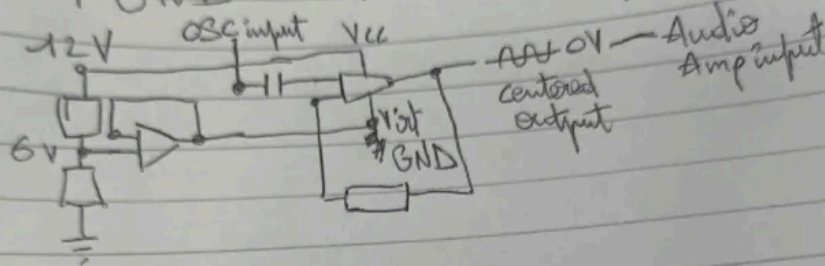


I then looked at the datasheet and saw that the chip can handle 12V.

And thought I could also generate a virtual gnd from the 12V supply so that I can then morph the signal to my audio amp's requirement through an inverting amplifier combined with an ac coupling capacitor.

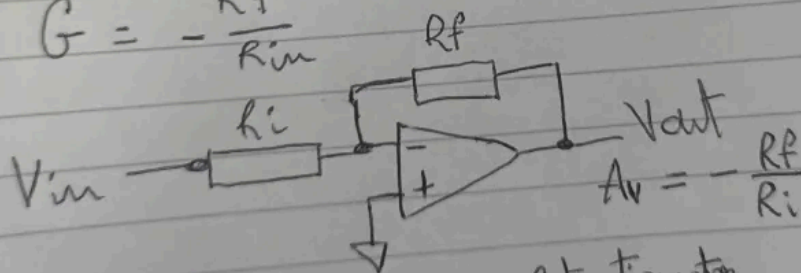
Output Level and scaling + Volume control?

- V_{out} GND at 6V + AC Coupling needed



- Set Pre amp Gain factor
Attenuation = inverting

$$G = -\frac{R_f}{R_{in}}$$



Potentiometer

But make R_f a ~~variable~~ ^{Potentiometer} so can control volume, make sure that make ∇ setting, so smallest R_f , can't go below certain level, by series R .

Once all of this was done, I drafted a quick inverting + virt gnd + ac coupling circuit idea which will be ready for testing, I also wrote down the maths for attenuation and circuit topology.

Schematic Design:

Part Organisation:

From my parts drawer I have the following components available:

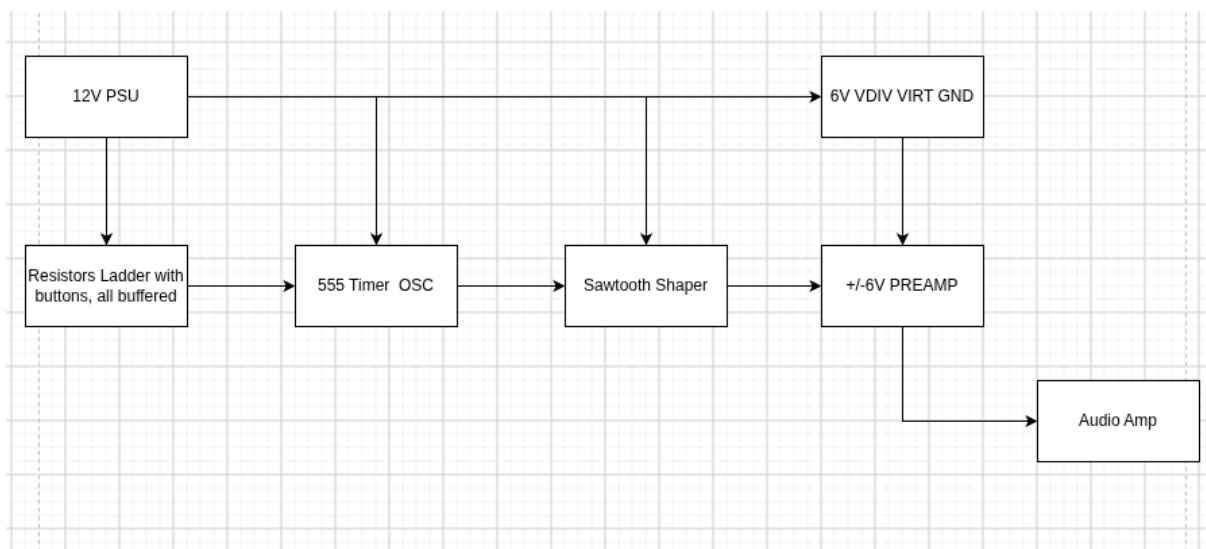
NE5532ADR, I will use this chip for the pre amp.

https://www.ti.com/lit/ds/symlink/ne5532.pdf?ts=1787995293486&ref_url=https%253A%252F%252Fcz.mouser.com%252F

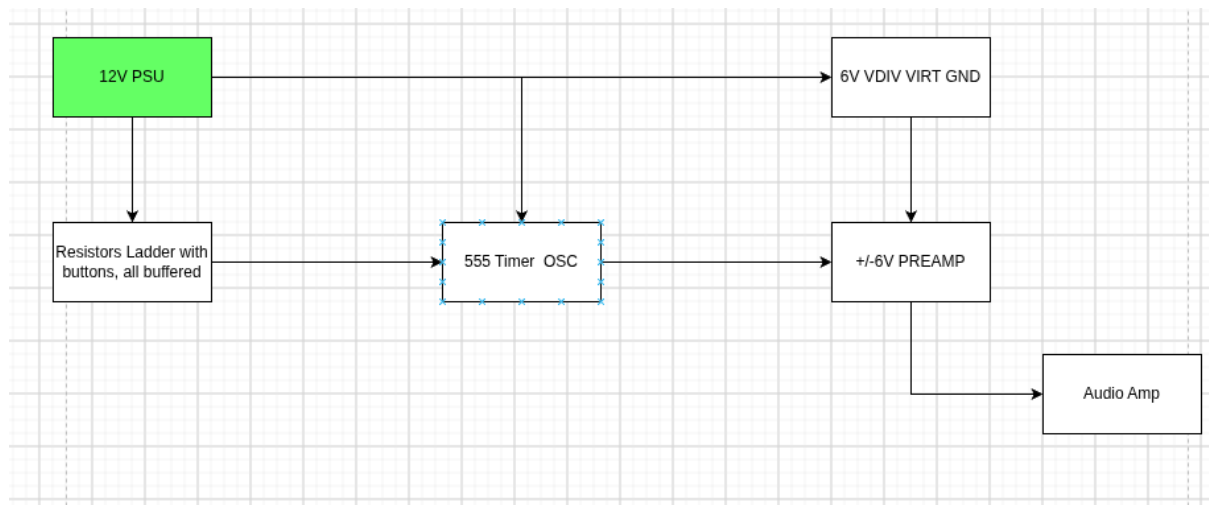
TL072HIDDFR, I will use this chip for the Virt GND

buffer. https://www.ti.com/lit/ds/symlink/tl071h.pdf?ts=1788006096240&ref_url=https%253A%252F%252Fwww.mouser.de%252F

All sorts of passives,
NE555.

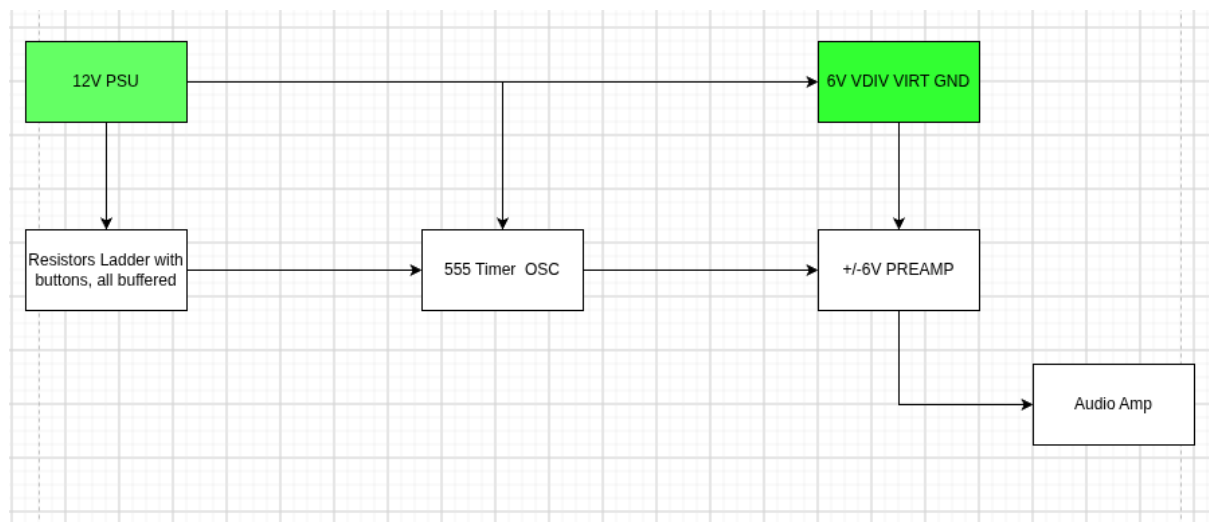


12V PSU:



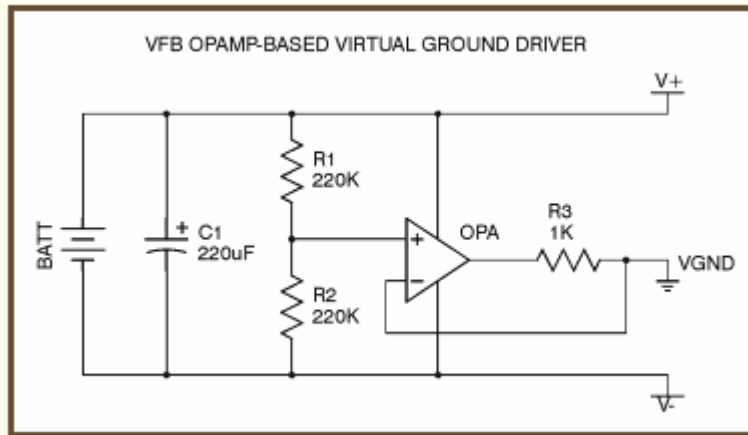
For now this is just a 12V supply from mains AC, nothing to do here,

6V Vdiv Virt GND:



This is about taking the 12V rail, having a Voltage divider network and then buffering the mid point to be used as a Virtual mid point ground,

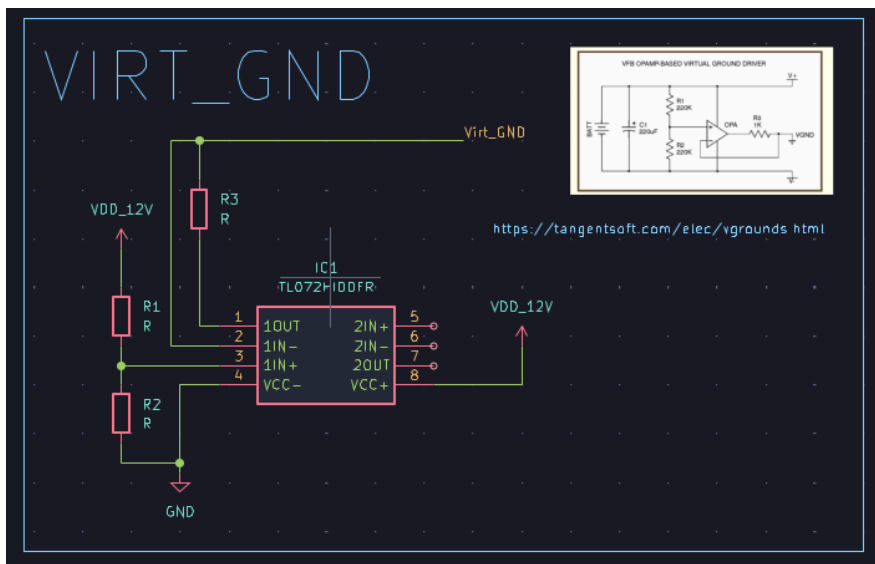
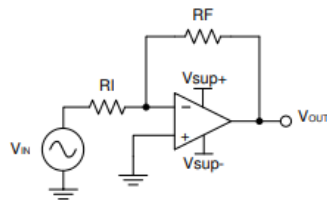
From Ref5:



This is to be done using the TL072HIDDFR, which according to the typical application can be treated as any opamp:

8.2 Typical Applications

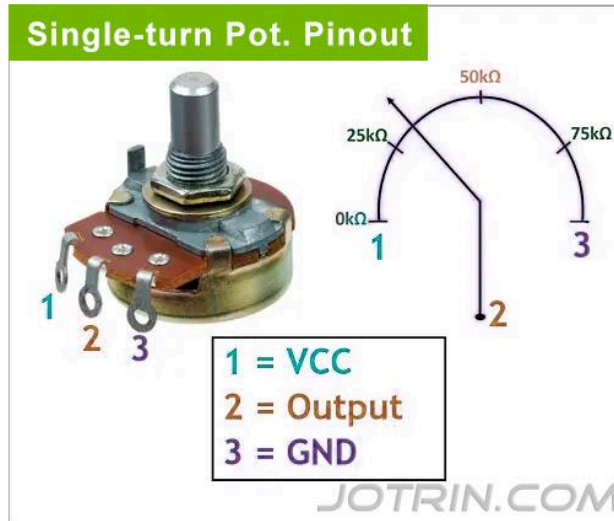
8.2.1 Inverting Amplifier



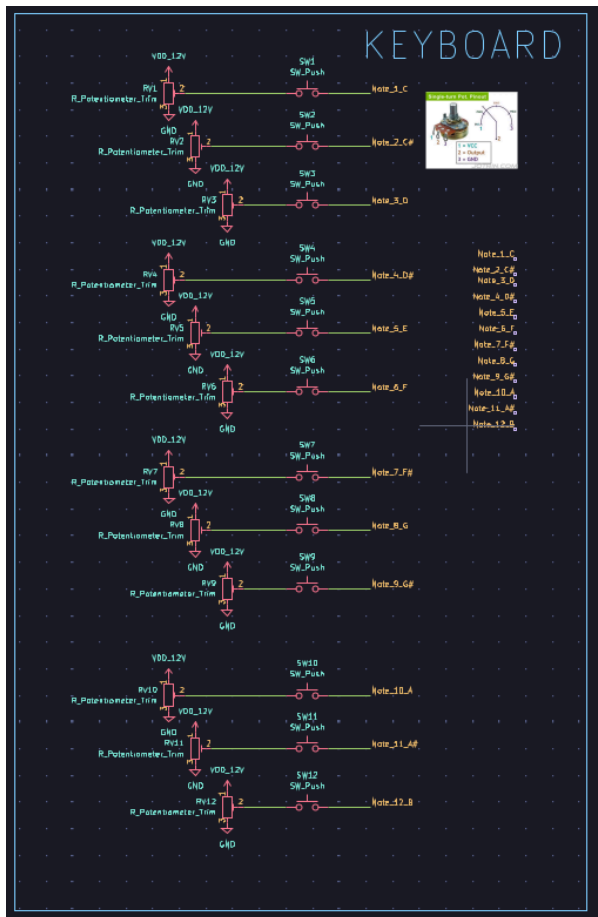
Resistors Ladder with Buttons + Buffered:

For this we need to use trimmers so that we can tune the instrument, as I am bypassing the maths for efficiency sake I will be setting an arbitrary trimmer value on the breadboard and seeing if it works then just adjust later if needed.

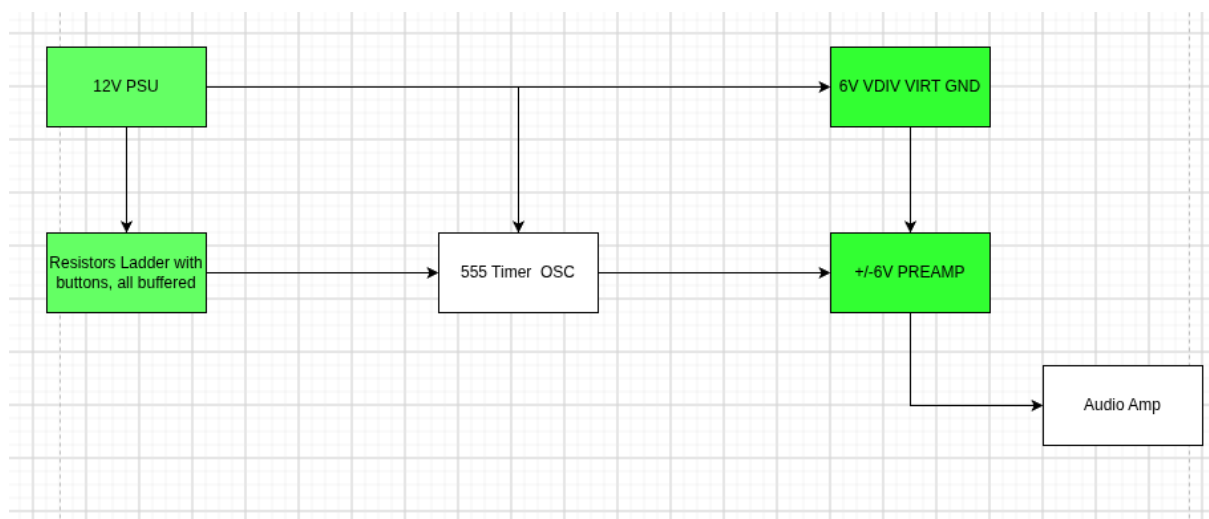
From Ref6, this is how a potentiometer works:



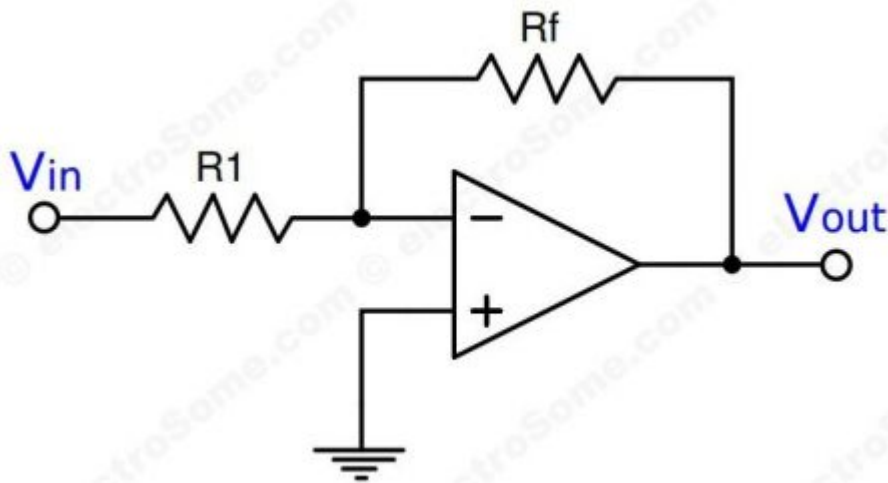
So you have the input go to pin 1, then the follow up goes to pin2 and then GND to pin 3. This creates a voltage divider, which means that pin 2 will output a variable voltage set to wherever you have the point, we don't actually need a cascading ladder which would induce issues such as adjusting the first trimmer would cascade the values of the next but here instead we can have each note be its own trimmer, which makes the job much simpler.



PRE_AMP:



For this we just need an OPAMP, in this case the NE5532 in an inverting topology:



Gain is then: $\text{Gain} = - (R_f / R_1)$

So if we predict the 555 timer to output 5V and we want that to become 0.4V:

$$0.4 / 5 = 0.08,$$

Setting R_f to 100k for reference we get:

$$-0.08 = (100k / R_1) \text{ Which transforms into: } R_1 = -(100k / (-0.08))$$

This would mean $R_1 = 1,250,000$, which is too big of a value for my liking, so I will change R_f to 8k and now R_1 is 100k, clean numbers.

The negative sign here indicates that the signal flips polarity.

However I didn't account for the fact that this is about comparing 0.4Vrms with whatever the rms of 5V is, so they need to both be rms not one in pkpk and one in rms,

Rms formulae:

RMS Voltage Equation

$$V_{\text{RMS}} = V_{\text{pk}} \frac{1}{\sqrt{2}} = V_{\text{pk}} \times 0.7071$$

So $5V_{\text{pk}} = \sim 3.5V_{\text{rms}}$

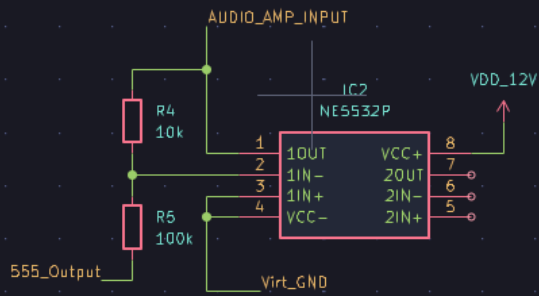
So now the attenuation is $0.4 / 3.5 = \sim 0.1$,

$$R_1 = -(10k / (-0.1))$$

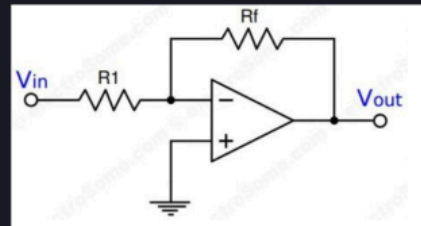
$$R_1 = 100k$$

I rounded some figures as it doesn't need to be completely accurate at this stage and there will be some breadboarding done later too

PRE-AMP



VIRT GND USED HERE!



Rms formulae:

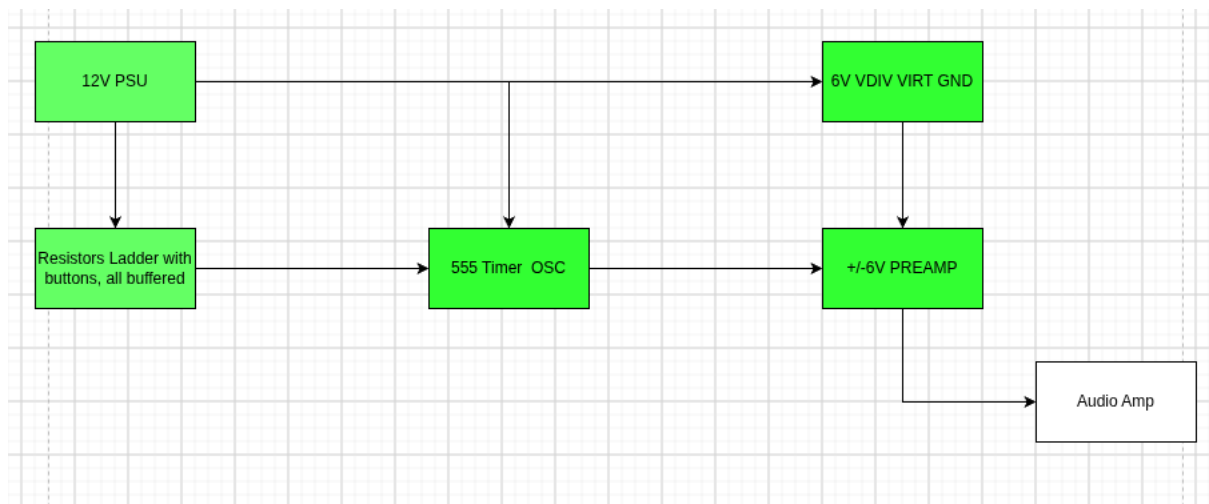
RMS Voltage Equation

$$V_{RMS} = V_{pk} \frac{1}{\sqrt{2}} = V_{pk} \times 0.7071$$

So 5Vpk = ~ 3.5Vrms

So now the attenuation is $0.4 / 3.5 = -0.1$,
 $R1 = -(10k / (-0.1))$
 $R1 = 100k$

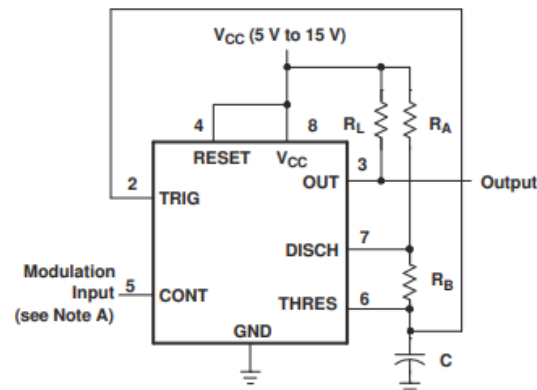
555 OSC:



For this I will simply follow the typical application notes of this chip:

7.2.3 Pulse-Position Modulation

As shown in Figure 7-5, any of these timers can be used as a pulse-position modulator. This application modulates the threshold voltage and, thereby, the time delay, of a free-running oscillator. Figure 7-6 shows a triangular-wave modulation signal for such a circuit; however, any wave shape can be used.



Pin numbers shown are for the D, JG, P, PS, and PW packages.

NOTE A: The modulating signal can be direct or capacitively coupled to CONT. For direct coupling, the effects of modulation source voltage and impedance on the bias of the timer should be considered.

7.2.3.2 Detailed Design Procedure

The nominal output frequency and duty cycle are determined using the formulas in Section 6.3.2. R_L improves V_{OH} , but R_L is not required for TTL compatibility.

7.2.3.3 Application Curve

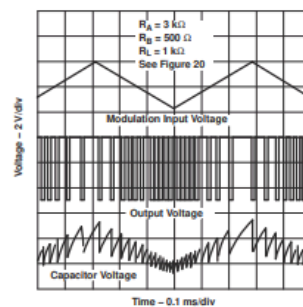
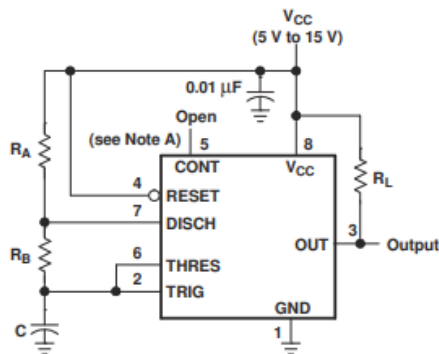


Figure 7-6. Pulse-Position-Modulation Waveforms

And the maths is detailed here:

6.3.2 Astable Operation

Figure 6-5 shows that adding a second resistor, R_B , to the circuit of Figure 6-2 and connecting the trigger input to the threshold input causes the timer to self-trigger and run as a multivibrator. Capacitor C charges through R_A and R_B and then discharges through R_B only. Therefore, the duty cycle is controlled by the values of R_A and R_B .



Pin numbers shown are for the D, JG, P, PS, and PW packages.
NOTE A: Decoupling CONT voltage to ground with a capacitor can improve operation. This should be evaluated for individual applications.

Figure 6-5. Circuit for Astable Operation

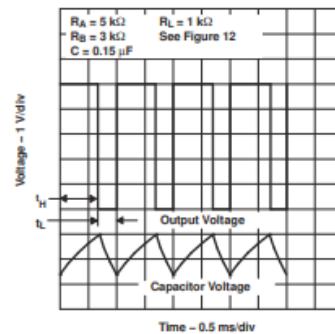


Figure 6-6. Typical Astable Waveforms

This astable connection results in capacitor C charging and discharging between the threshold-voltage level ($\cong 0.67 \times V_{CC}$) and the trigger-voltage level ($\cong 0.33 \times V_{CC}$). As in the mono-stable circuit, charge and discharge times (and, therefore, the frequency and duty cycle) are independent of the supply voltage. To reduce distortion, use at maximum frequency of 100kHz or below. If higher-frequency operation is required, consider using the [TLC555 CMOS Timer](#) instead.

Figure 6-6 shows typical waveforms generated during astable operation. The output high-level duration t_H and low-level duration t_L are calculated as follows:

$$t_H \cong 0.693 \times (R_A + R_B) \times C \quad (1)$$

$$t_L \cong 0.693 \times R_B \times C \quad (2)$$

Other useful relationships for period, frequency, and driver-referred and waveform-referred duty cycle are calculated as follows:

$$T = t_H + t_L \cong 0.693 \times (R_A + 2R_B) \times C \quad (3)$$

$$f = \frac{1}{T} \cong \frac{1.44}{(R_A + 2R_B) \times C} \quad (4)$$

$$\text{Output driver duty cycle} = \frac{t_L}{T} \cong \frac{R_B}{R_A + 2R_B} \quad (5)$$

$$\text{Output waveform duty cycle} = \frac{t_H}{T} \cong 1 - \frac{R_B}{R_A + 2R_B} = \frac{R_A + R_B}{R_A + 2R_B} \quad (6)$$

So from my understanding frequency = $1.44 / (R_A + 2R_B) \times C$

However for PPM configuration:

7.2.3.1 Design Requirements

Both dc- and ac-coupled **modulation input** changes the upper and lower voltage thresholds for the timing capacitor. Both frequency and duty cycle vary with the modulation voltage.

The modulation input changes the voltage thresholds which then changes the frequency as the charge/discharge will happen at a different rate, so this seems like a bit of a rabbit hole and I think the most efficient way to solve this is to follow the typical application graph values of:

$R_A = 3\text{ k ohms}$, $R_B = 500\text{ ohms}$, $R_L = 1\text{ k ohms}$

Then on a breadboard setup just one pot trimmer Vdiv input to the modulation input, figure out what R values give me C3 through the 12 notes up to B3, take notes of them and then select the trimmers based on this experiment.

555_OSC

Using astable application note to choose a value, can then adjust based on testing.

7.2.3 Pulse-Position Modulation
As shown in Figure 7-6, any of these timers can be used as a pulse-position modulator. This application modulates the threshold voltage and, thereby, the time delay of a free-running oscillator. Figure 7-6 shows a triangular-wave modulation signal for such a circuit; however, any wave shape can be used.

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RA = 3k ohms, RB = 500 ohms, RL = 1k ohms

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Figure 6-5 shows that adding a second resistor, R_B , to the circuit of Figure 6-2 and connecting the trigger input to the threshold input causes the timer to self-trigger and run as a multivibrator. Capacitor C charges through R_A and R_B and then discharges through R_B only. Therefore, the duty cycle is controlled by the values of R_A and R_B .

Figure 6-4. Typical Astable Waveforms

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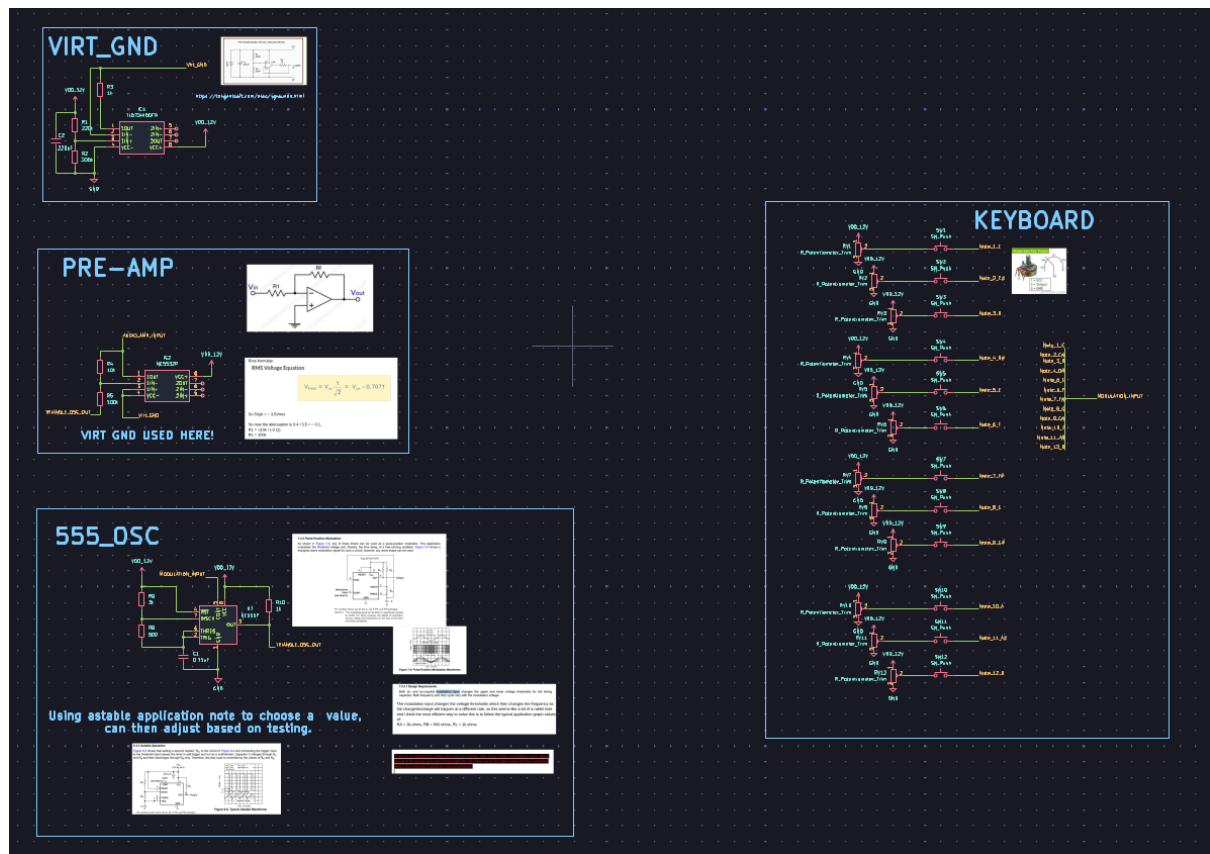
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Figure 6-4. Typical Astable Waveforms

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Both dc- and ac-coupled ~~modulation~~ charges the upper and lower voltage thresholds for the timing capacitor. Both frequency and duty cycle vary with the modulation voltage.
The modulation input charges the voltage thresholds which then changes the frequency as the charged/discharged will happen at a different rate, so this seems like a bit of a rabbit hole and I think the most efficient way to solve this is to follow the typical application graph values of.
RA = 3k ohms, RB = 500 ohms, RL = 1k ohms

Tidy up and Connect to Audio Amp:



Design Error, Research Rectification, Learning:

In/Out Impedance:

I then realised through another build that potentially the virtual ground and pre-amp aren't actually needed,

Maybe just a Resistor divider would be sufficient for converting the 12V PWM output to the input required for the audio amp,

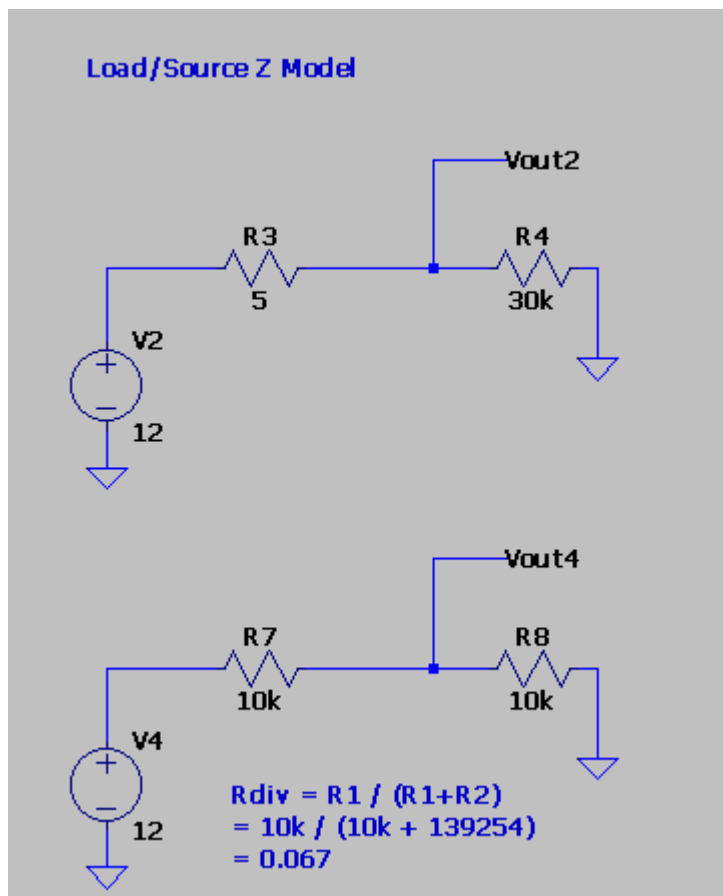
Then an AC Coupling cap is used to remove the DC offset and center the signal around 0V,

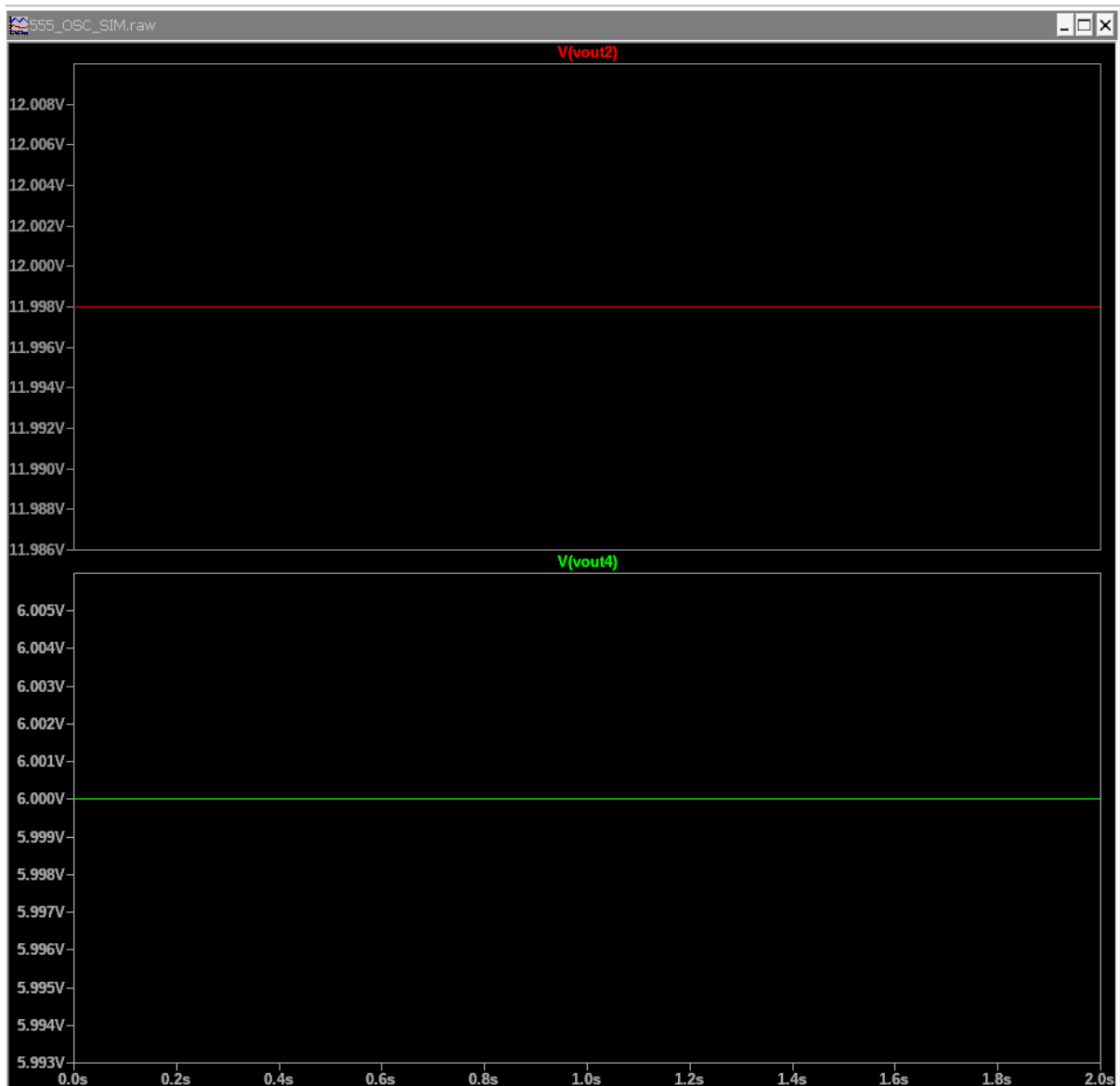
So I came to this decision by learning about Load and Source impedance and it's ratios relationship with the delivered voltage:

Load Z / Source Z = Load to Source Z ratio.

$V_L = V_S * (R_L / R_S + R_L)$ -> so the resistors ratio define the percentage of V that is delivered.

Here's a model of the ratio and its relationship with delivered voltage:





As you can see, having a good load to source ratio means that more voltage is present at the load, if the load = source then you would lose half of the voltage at the load, this is crucial for high precision sensing equipment as the sample is already so small that we want to obtain as much of it as possible inside of the sensing equipment to amplify it.

TPA3116 input impedance and cutoff calculations:

Feature Description (continued)

Table 1. Gain and Master/Slave

MASTER / SLAVE MODE	GAIN	R1 (to GND) ⁽¹⁾	R2 (to GVDD) ⁽¹⁾	INPUT IMPEDANCE
Master	20 dB	5.6 kΩ	OPEN	60 kΩ
Master	26 dB	20 kΩ	100 kΩ	30 kΩ
Master	32 dB	39 kΩ	100 kΩ	15 kΩ
Master	36 dB	47 kΩ	75 kΩ	9 kΩ
Slave	20 dB	51 kΩ	51 kΩ	60 kΩ
Slave	26 dB	75 kΩ	47 kΩ	30 kΩ
Slave	32 dB	100 kΩ	39 kΩ	15 kΩ
Slave	36 dB	100 kΩ	16 kΩ	9 kΩ

7.3.2 Input Impedance

The TPA31xxD2 family input stage is a fully differential input stage and the input impedance changes with the gain setting from 9 kΩ at 36 dB gain to 60 kΩ at 20 dB gain. Table 1 lists the values from min to max gain. The tolerance of the input resistor value is $\pm 20\%$ so the minimum value will be higher than 7.2 kΩ. The inputs need to be AC-coupled to minimize the output dc-offset and ensure correct ramping of the output voltages during power-ON and power-OFF. The input ac-coupling capacitor together with the input impedance forms a high-pass filter with the following cut-off frequency:

$$f = \frac{1}{2\pi Z_i C_i} \quad (1)$$

If a flat bass response is required down to 20 Hz the recommended cut-off frequency is a tenth of that, 2 Hz. Table 2 lists the recommended ac-couplings capacitors for each gain step. If a -3 dB is accepted at 20 Hz 10 times lower capacitors can be used – for example, a 1 μF can be used.

Table 2. Recommended Input AC-Coupling Capacitors

GAIN	INPUT IMPEDANCE	INPUT CAPACITANCE	HIGH-PASS FILTER
20 dB	60 kΩ	1.5 μF	1.8 Hz
26 dB	30 kΩ	3.3 μF	1.6 Hz
32 dB	15 kΩ	5.6 μF	1.9 Hz
36 dB	9 kΩ	10 μF	1.8 Hz

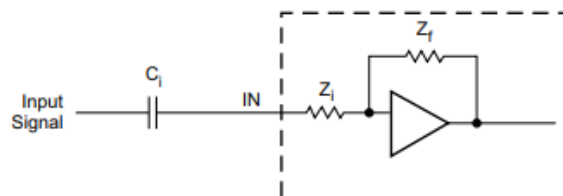
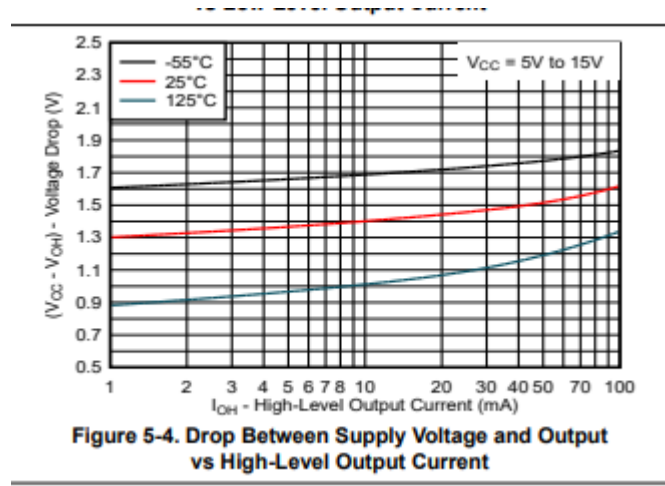


Figure 28. Input Impedance

The input capacitors used should be a type with low leakage, like quality electrolytic, tantalum or ceramic. If a polarized type is used the positive connection should face the input pins which are biased to 3 Vdc.

555 Timer output impedance:



This isn't a perfect science but estimating we can see that for 25 degrees C the voltage drop goes from 1.3V to 1.5V in 40mA, that is a 0.2V change in 40mA, $R = 0.2/0.04 = 5$ ohms

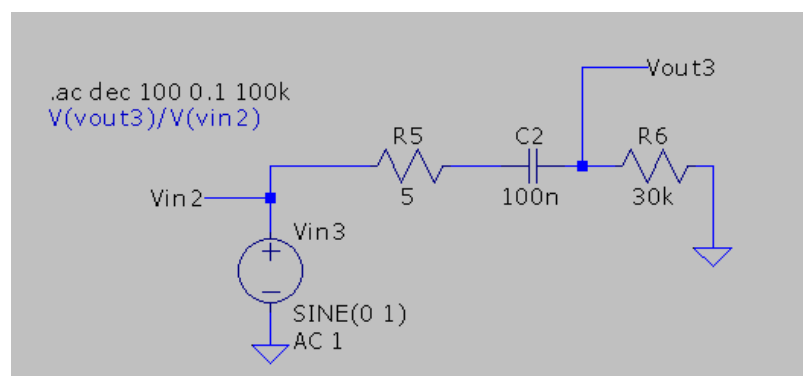
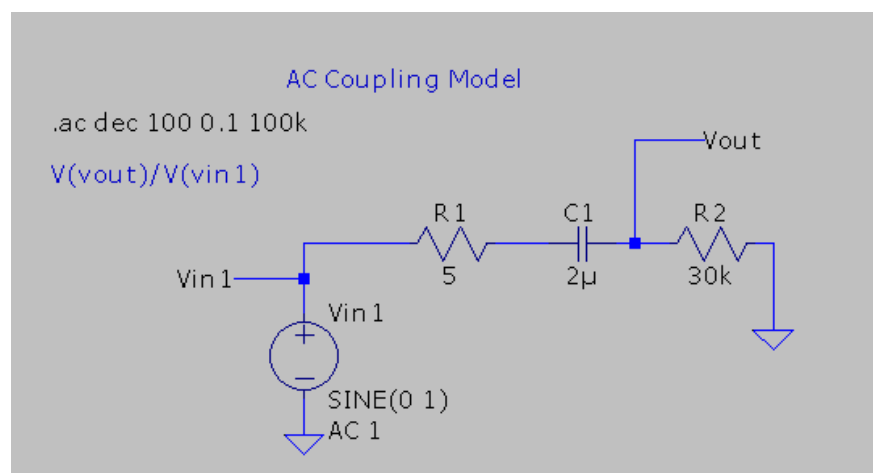
AC Coupling:

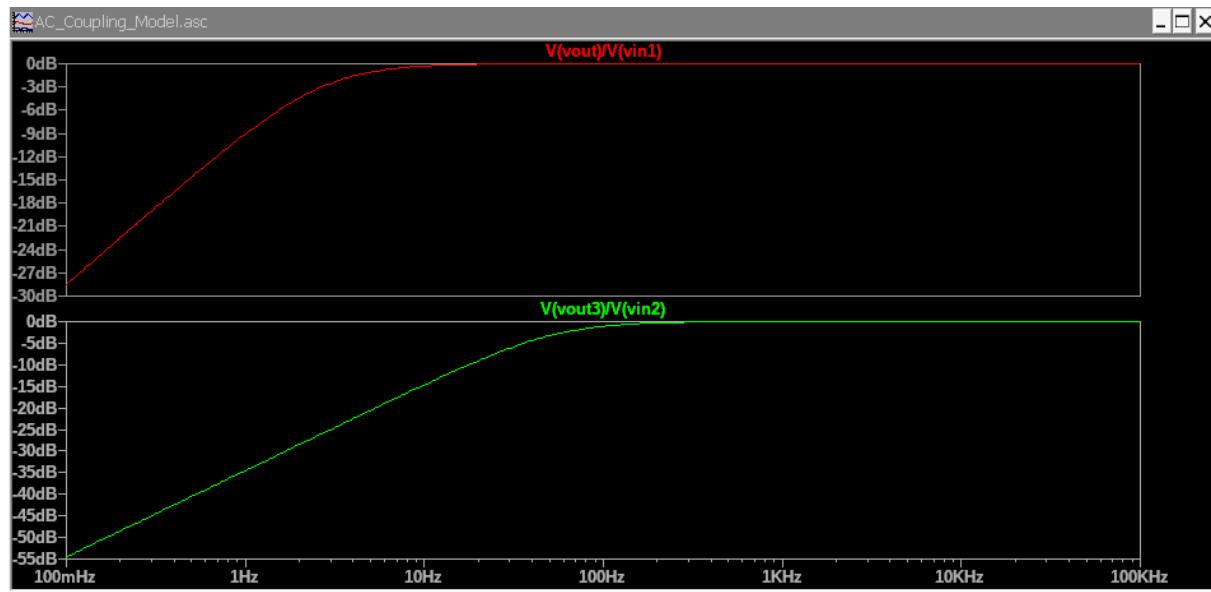
I also got to learn about AC coupling, and how a coupling cap in combination with the load impedance acts as a high pass filter configuration, a higher capacitance cap is thus then resulting in a better preservation of the lower frequency constituents of an audio waveform,

Maths:

$$F_c = 1/(2\pi RC)$$

Model:





555 PPM Frequency calculation:

I also thought I'd seize this opportunity for a change to do the frequency calculation properly by:

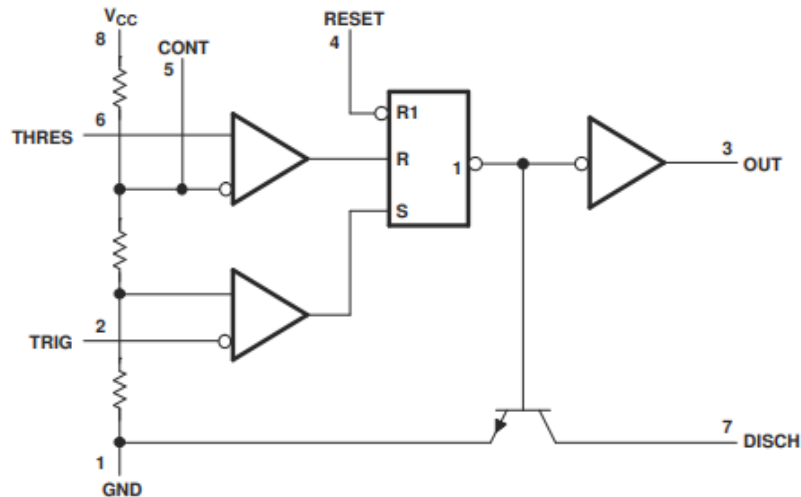
- Figuring out the mathematics for it from the datasheet,
- Plotting breadboard test values and comparing it with the maths,
- Deciding whether I make the buttons a stable PCB or a vero board with potentiometers, this will depend on the stability of the system with respect to temperature and perhaps other unknown impactful variables.

555 PPM F Mathematics:

6 Detailed Description

6.1 Overview

The Nx555 or Sx555 is a precision timing device for general-purpose timing applications from 10 μ s to hours or from < 1mHz to 100kHz. In the time-delay or monostable mode of operation, the timed interval is controlled by a single external resistor and capacitor network. In the astable mode of operation, the frequency and duty cycle can be controlled independently with two external resistors and a single external capacitor. Maximum output sink and discharge sink current are greater for higher V_{CC} and less for lower V_{CC} .

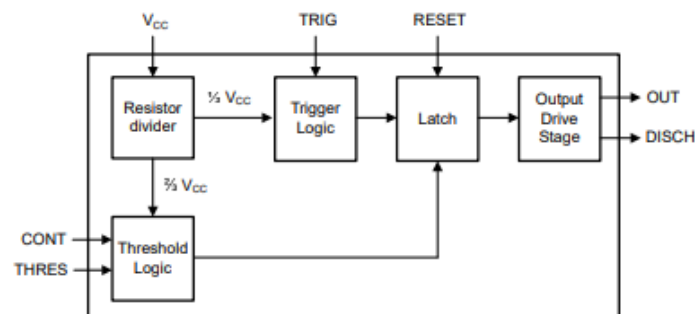


Note: Pin numbers shown are for the D, JG, P, PS, and PW packages.

Note: RESET can override TRIG, which can override THRES.

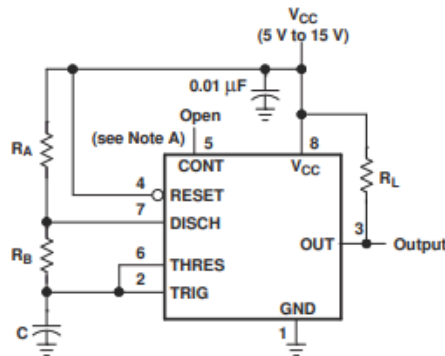
Figure 6-1. Simplified Schematic

6.2 Functional Block Diagram



6.3.2 Astable Operation

Figure 6-5 shows that adding a second resistor, R_B , to the circuit of Figure 6-2 and connecting the trigger input to the threshold input causes the timer to self-trigger and run as a multivibrator. Capacitor C charges through R_A and R_B and then discharges through R_B only. Therefore, the duty cycle is controlled by the values of R_A and R_B .



Pin numbers shown are for the D, JG, P, PS, and PW packages.
NOTE A: Decoupling CONT voltage to ground with a capacitor can improve operation. This should be evaluated for individual applications.

Figure 6-5. Circuit for Astable Operation

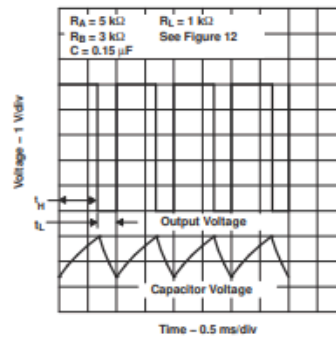


Figure 6-6. Typical Astable Waveforms

This astable connection results in capacitor C charging and discharging between the threshold-voltage level ($\approx 0.67 \times V_{CC}$) and the trigger-voltage level ($\approx 0.33 \times V_{CC}$). As in the mono-stable circuit, charge and discharge times (and, therefore, the frequency and duty cycle) are independent of the supply voltage. To reduce distortion, use at maximum frequency of 100kHz or below. If higher-frequency operation is required, consider using the [TLC555 CMOS Timer](#) instead.

Figure 6-6 shows typical waveforms generated during astable operation. The output high-level duration t_H and low-level duration t_L are calculated as follows:

$$t_H \approx 0.693 \times (R_A + R_B) \times C \quad (1)$$

$$t_L \approx 0.693 \times R_B \times C \quad (2)$$

Other useful relationships for period, frequency, and driver-referred and waveform-referred duty cycle are calculated as follows:

$$T = t_H + t_L \approx 0.693 \times (R_A + 2R_B) \times C \quad (3)$$

$$f = \frac{1}{T} \approx \frac{1.44}{(R_A + 2R_B) \times C} \quad (4)$$

$$\text{Output driver duty cycle} = \frac{t_L}{T} \approx \frac{R_B}{R_A + 2R_B} \quad (5)$$

$$\text{Output waveform duty cycle} = \frac{t_H}{T} \approx 1 - \frac{R_B}{R_A + 2R_B} = \frac{R_A + R_B}{R_A + 2R_B} \quad (6)$$

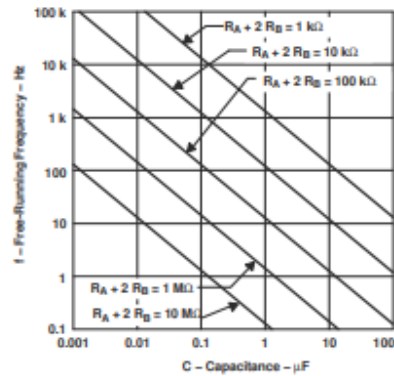
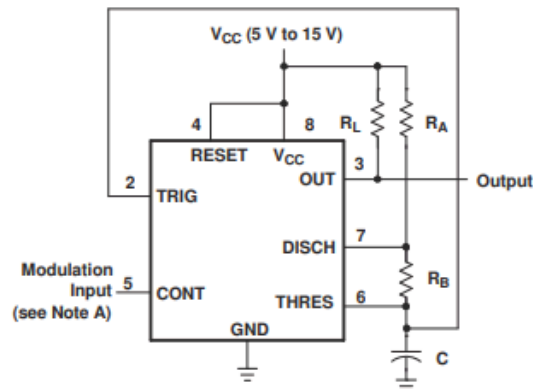


Figure 6-7. Free-Running Frequency

7.2.3 Pulse-Position Modulation

As shown in Figure 7-5, any of these timers can be used as a pulse-position modulator. This application modulates the threshold voltage and, thereby, the time delay, of a free-running oscillator. Figure 7-6 shows a triangular-wave modulation signal for such a circuit; however, any wave shape can be used.



Pin numbers shown are for the D, JG, P, PS, and PW packages.

NOTE A: The modulating signal can be direct or capacitively coupled to CONT. For direct coupling, the effects of modulation source voltage and impedance on the bias of the timer should be considered.

The note here mentions that the effects modulation of the source voltage and impedance of the timer should be considered.

7.2.3.1 Design Requirements

Both dc- and ac-coupled modulation input changes the upper and lower voltage thresholds for the timing capacitor. Both frequency and duty cycle vary with the modulation voltage.

Equation for charging an RC circuit

General relationship

$$V_C = V_S \left[1 - e \left(\frac{-t}{\tau} \right) \right] \quad (38)$$

Where

V_C = voltage across the capacitor at any instant in time (t)

V_S = the source voltage charging the RC circuit

t = time in seconds

τ = RC, the time constant for charging and discharging capacitors

Graphing **Equation 37** produces the capacitor charging curve below. Note that the capacitor is 99.3% charged at five time constants. It is common practice to consider this *fully charged*.

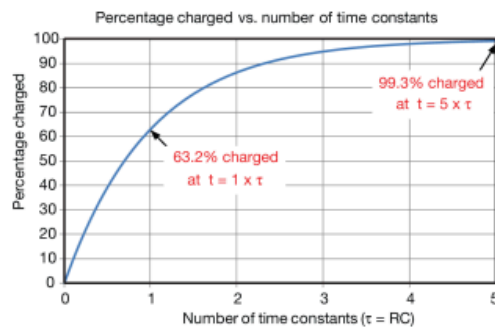


Figure 13. RC charge curve

General Relationship

$$V_C = V_i \left[e \left(\frac{-t}{\tau} \right) \right] \quad (39)$$

Where

V_C = voltage across the capacitor at any instant in time (t)

V_i = the initial voltage of the capacitor at t = 0s

t = time in seconds

τ = RC, the time constant for charging and discharging capacitors

Graphing **Equation 39** produces the capacitor discharge curve below. Note that the capacitor is discharged to 0.7% at five time constants. It is common practice to consider this *fully discharged*.

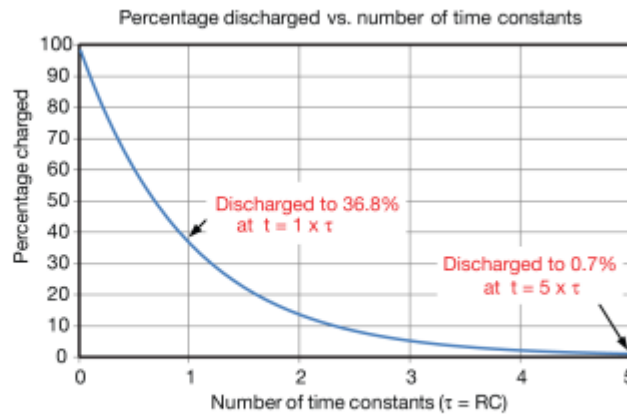


Figure 14. *RC discharge curve*

7.2.3.2 Detailed Design Procedure

The nominal output frequency and duty cycle are determined using the formulas in [Section 6.3.2](#). R_L improves V_{OH} , but R_L is not required for TTL compatibility.

7.2.3.3 Application Curve

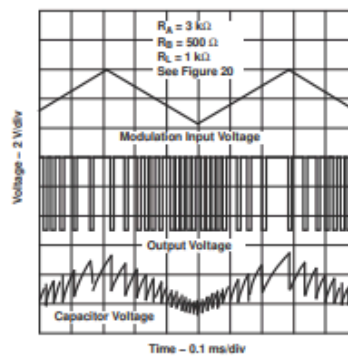
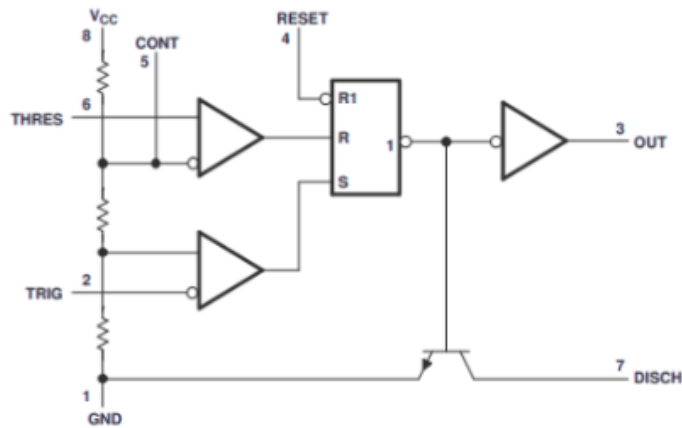


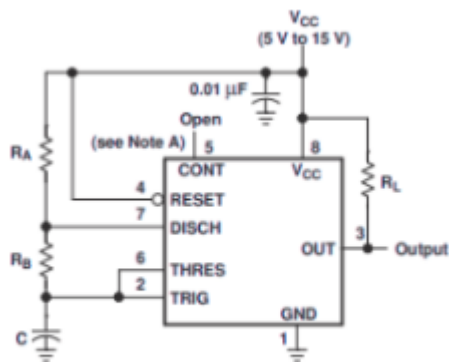
Figure 7-6. *Pulse-Position-Modulation Waveforms*

If we combine the astable circuit diagram where we can see the passives like discharge cap and it's relevant resistors with the Overview diagram then we can get a good idea of the overall picture:



Note: Pin numbers shown are for the D, JG, P, PS, and PW packages.

Note: RESET can override TRIG, which can override THRES.



Pin numbers shown are for the D, JG, P, PS, and PW packages.

NOTE A: Decoupling CONT voltage to ground with a capacitor can improve operation. This should be evaluated for individual

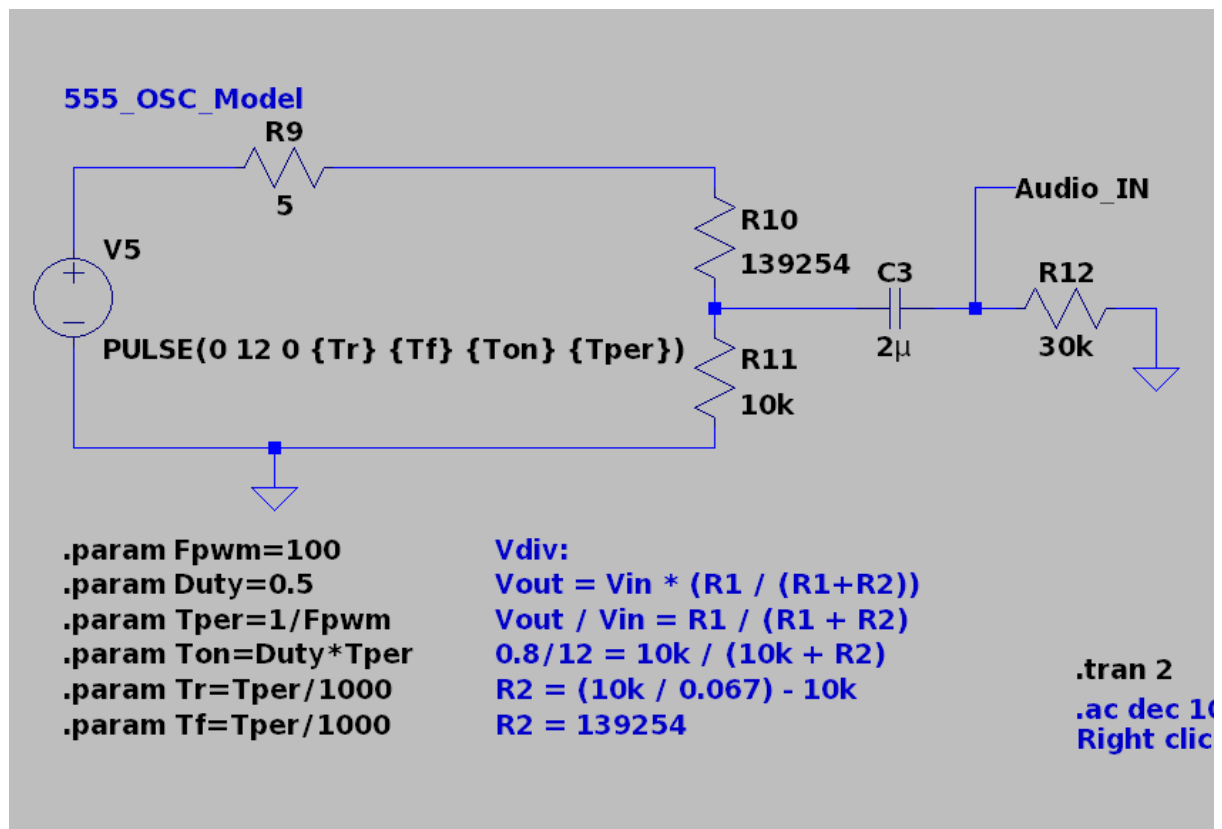
There is an SR latch in the overview which needs analyzing, but we can see that DISCH would get pulled to GND level when the SR triggers and we know that that triggering is due to the two different threshold point being reached,

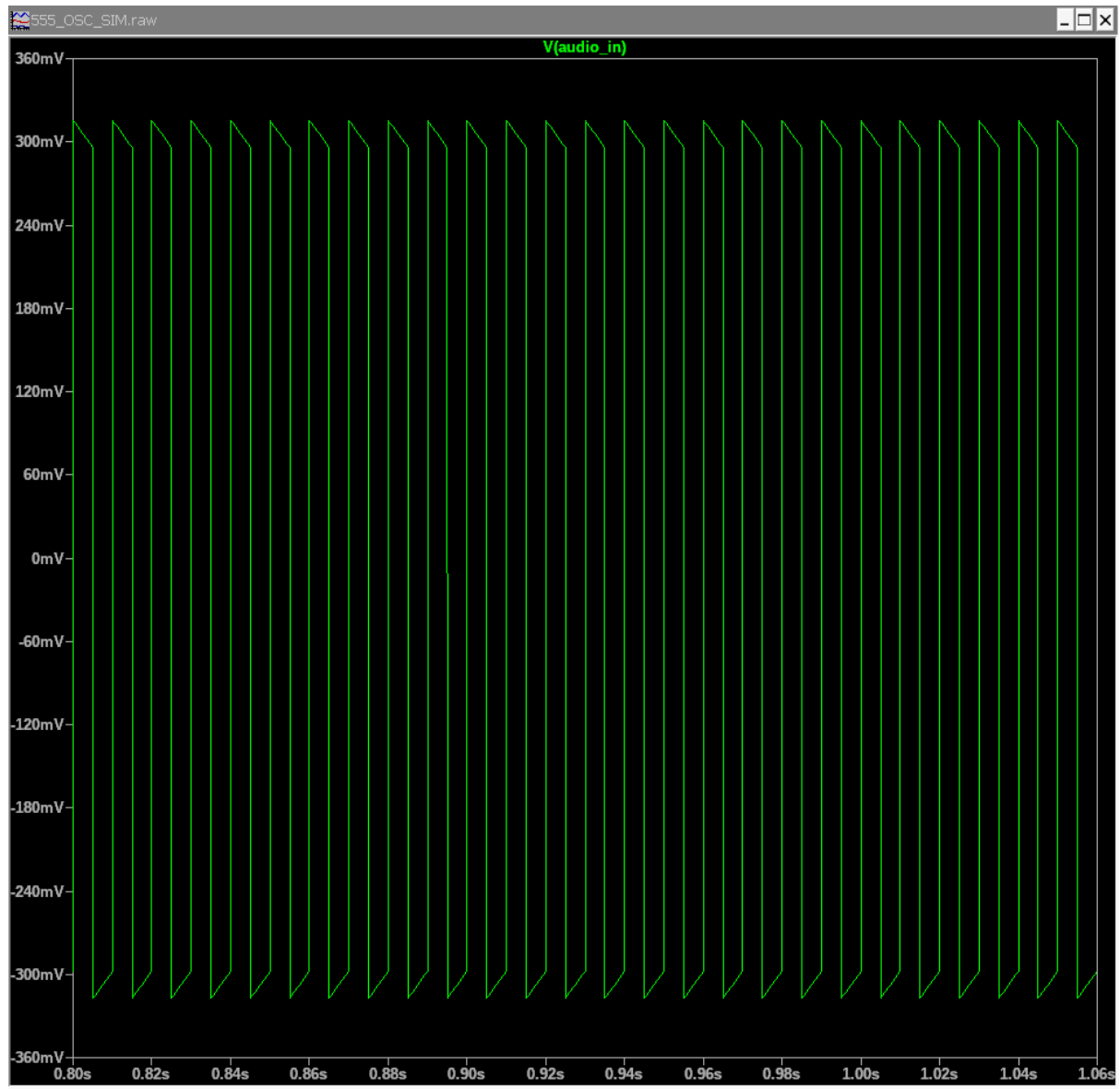
If we look at the astable circuit we can see that discharge is controlling the point above the Threshold and trigger points and englobes the caps, thus if DISCH is set to GND level then the cap would stop being powered and would discharge into the grounded load

555 PPM F Breadboard Testing:

555 PPM F Design Decision:

Full circuit simulation with new concept:





New schematic design:

Breadboard prototyping:

Testing Results:

- Built using slightly different passive values,
- Forgot about the AC coupling on the schematic, added 100nF on breadboard for testing,
- It seems the pulse to pulse gap changes based on potentiometer settings but the pulses do look weird,
- Can't easily test with scope due to Earth induced GND bridging,

Some measurements:

I tested every section block by block in isolation,

Virt GND, CORRECT, 6V is outputted from the chip, to be fed as GND for the attenuation circuit, and we see both + and - 6V.

Buttons, CORRECT, based on current circuit, when no button is pressed $V = 0$, when I press a button then I get the voltage from the VDIV

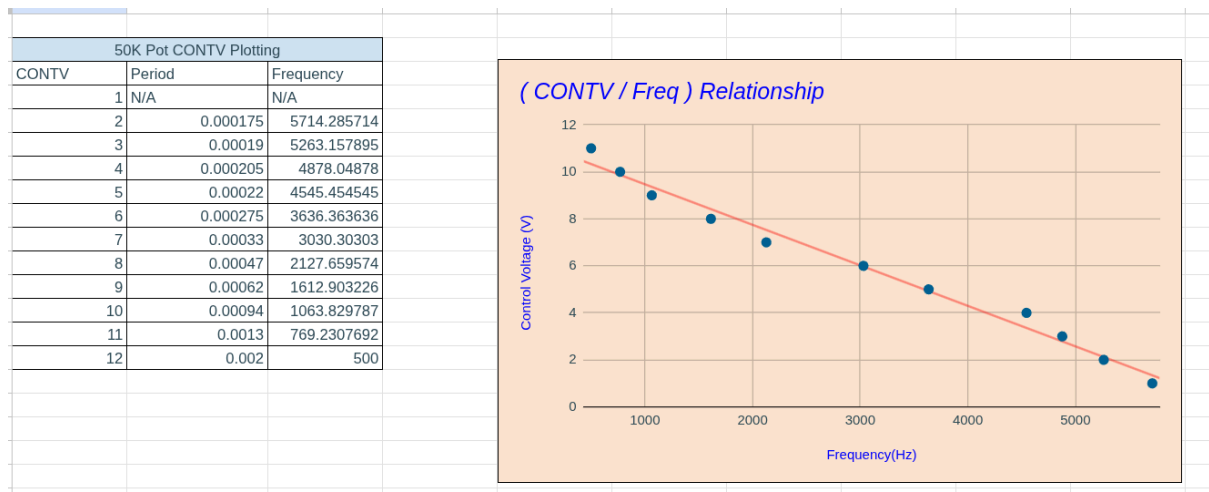
PRE-AMP, ISSUE, here when trying to scope the output, virt gnd collapses and the output becomes centered around the wrong value, however it seems the circuit is behaving correctly. HOWEVER, when scoping using a non grounded probe, the behaviour seems correct, the pre amp outputted pulse is about 1.5pk, whereas the output of the 555 seems to be 12.5V, thus the attenuation is correct, I just assumed that the 555 would output 5Vpk and learnt that it actually outputs V_{cc} .

When powering the circuit with 5V, we see the correct behaviour of the 555 timer outputting a 5Vpk waveform and the preamp outputting a 1Vpk centered at 2.5V.

So now, fixes would involve:

- Correcting the attenuation to handle a 12Vpk waveform and attenuating it to 0.4Vrms
- Researching connecting this module with the audio amp module and in particular how GND should be handled.
- Add AC coupling components and whatever other issues may be found.

This is what I got after plotting the values:



This is the note frequency calculations, as we can see I can play from C5 to C6 for a full octave of notes by just going from 10V to 12V, this is sufficient for my use case so the next step is to calculate the the resistor values to get a 10/12V CONT input:

Notes:	$16.35 * 2^{(\text{semitone}/12)}$		
Semitone	Position	Frequency	
C0	0	16.35	
C#0	1	17.32222159	
C5	60	523.2	
C#5	61	554.311091	
D5	62	587.2721437	
D#5	63	622.1931626	
E5	64	659.1906933	
F5	65	698.3882117	
F#5	66	739.9165358	
G5	67	783.9142626	
G#5	68	830.5282304	
A5	69	879.9140089	
A#5	70	932.2364187	
B5	71	987.6700808	
C6	72	1046.4	
So my range of interest is 500Hz to 1050Hz			
Which is 12V to 10V Control Voltage			

However recalling the following from the datasheet:

Figure 6-6 shows typical waveforms generated during astable operation. The output high-level duration t_H and low-level duration t_L are calculated as follows:

$$t_H \cong 0.693 \times (R_A + R_B) \times C \quad (1)$$

$$t_L \cong 0.693 \times R_B \times C \quad (2)$$

Other useful relationships for period, frequency, and driver-referred and waveform-referred duty cycle are calculated as follows:

$$T = t_H + t_L \cong 0.693 \times (R_A + 2R_B) \times C \quad (3)$$

$$f = \frac{1}{T} \cong \frac{1.44}{(R_A + 2R_B) \times C} \quad (4)$$

$$\text{Output driver duty cycle} = \frac{t_L}{T} \cong \frac{R_B}{R_A + 2R_B} \quad (5)$$

$$\text{Output waveform duty cycle} = \frac{t_H}{T} \cong 1 - \frac{R_B}{R_A + 2R_B} = \frac{R_A + R_B}{R_A + 2R_B} \quad (6)$$

We can see that the frequency is relative to the R_A and R_B value, where increasing R_A and R_B would lead to a smaller frequency value, we can also see that the duty cycle is relative to ratio of R_A to R_B ,

So, keeping C as the current $0.15\mu\text{F}$ and aiming for a frequency of 16.35Hz so that we can reach lower octaves:

$$16.35 = 1.44 / ((R_{\text{factor}}) \times 0.15\mu)$$

$$\text{Where } R_{\text{factor}} = R_A + 2R_B$$

Thus:

$$R_{\text{factor}} = 1.44 / (16.35 \times 0.15\mu)$$

Finally:

$$R_A + 2R_B = 587k$$

Waveform referred duty cycle refers to the ratio of time the waveform is high in a time period: t_H / T .

Output driver duty cycle refers to the low: t_L / T

So aiming for a $x > 50\%$ duty cycle, since 50% wouldn't be possible without R_A being 0 .

We can try 51% :

$$0.51 = R_A + R_B / (587k)$$

$$0.51 \times 587k = R_A + R_B$$

$$299370 = R_A + R_B$$

If we set R_B as $100k$ for reference then R_A can be equal to $200k$.

In the future we can look at the effects of duty cycle sweeping on sounds.

However as 16.37Hz is $C0$ and I want something more mid range I will shift this to 130.8Hz or $C3$:

$$130.8 = 1.44 / ((R_{\text{factor}}) \times 0.15\mu)$$

Where $R_{factor} = R_A + 2R_B$

Thus:

$R_{factor} = 1.44 / (130.8 \times 0.15\mu)$

Finally:

$R_A + 2R_B = 73k394$

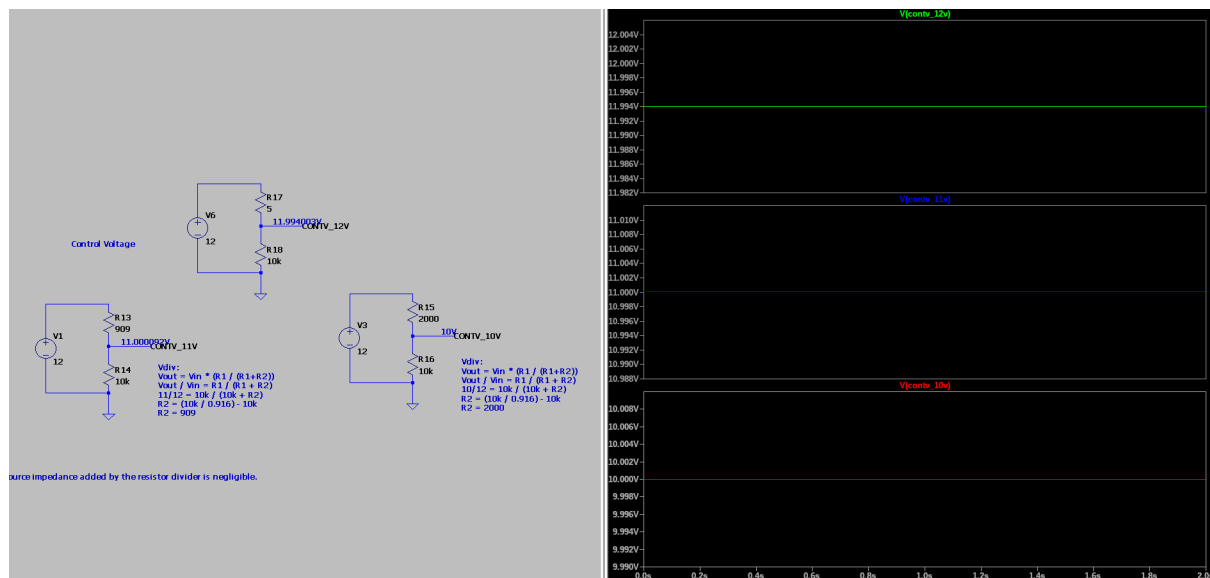
$0.51 = R_A + R_B / (73k394)$

$0.51 \times 73k394 = R_A + R_B$

$37k430.94 = R_A + R_B$

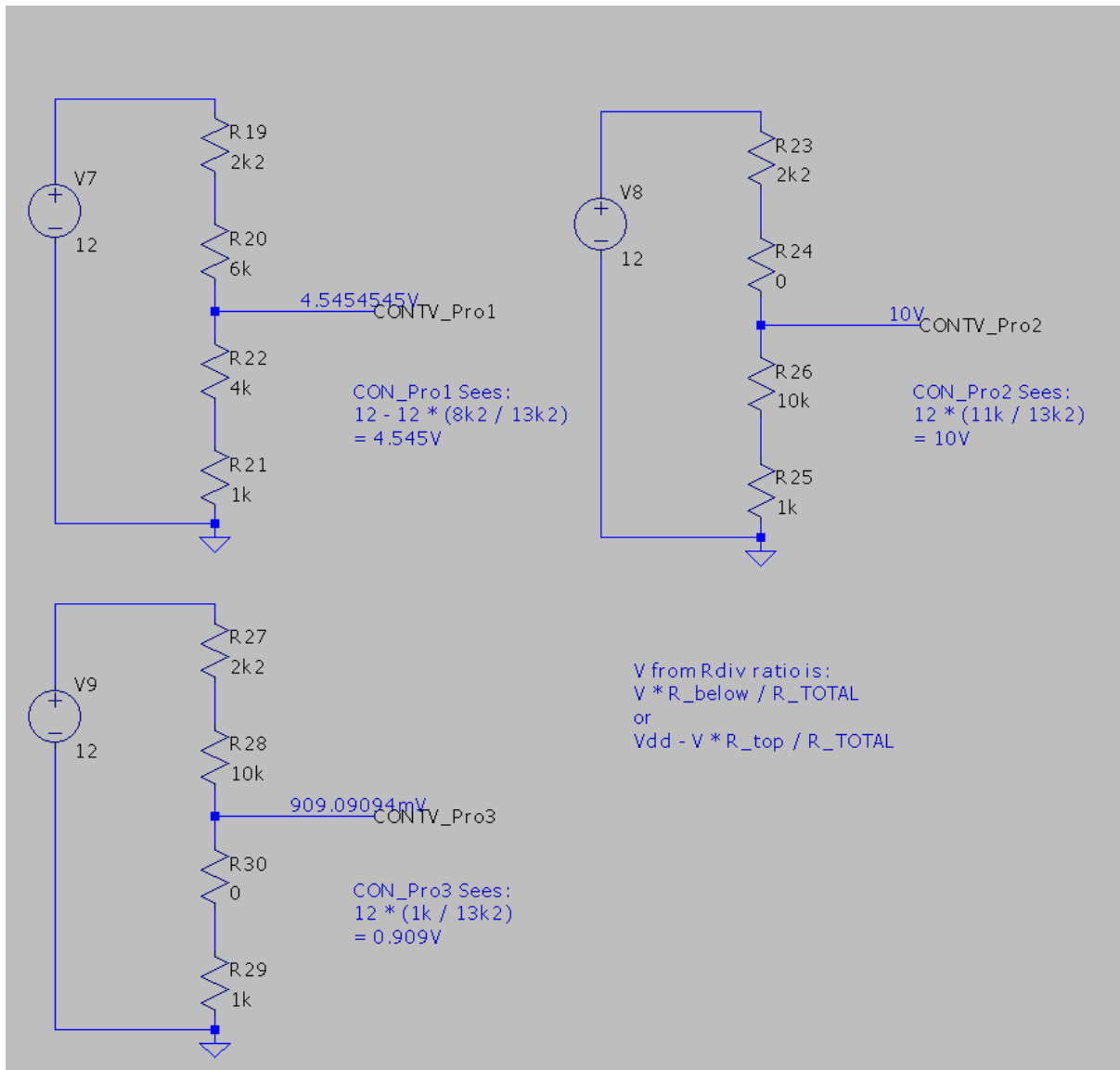
If we set R_B as 10k for reference then R_A can be equal to 27k.

I then simulated voltage dividers to ensure I was getting the maths correct to obtain voltages from 12V to 10V:



From this I figured a 2k pot in series with a 10k resistor where the midpoint between the 2 is gnd would be a good test to figure out the values needed for notes going from C5 to C6:

The CONT input which get controller by a pot trimmer needs Resistors at the top and bottom to limit the input so that it fits within 10V-2V to avoid driving it on the absolute edges which can cause some stalling.



The 555 timer is named after 5 series resistors in it's internal structure, this results in Pin5 (CONT) having a 3k3 impedance,

References:

References:

Ref1: notes and their frequencies

<https://www.idrumtune.com/ultimate-guide-to-musical-frequencies/>

Ref2: a complicated VCO + various waveform generators module. +/-12V
https://musicfromouterspace.com/analogsynth_new/VCO20090724REV0/VCO20090724REV0.html

Ref3: The basics of Oscillators and VCOs well explained step by step with circuits to be used +/-12V
<https://sound-au.com/project162.htm>

Ref4: 555 Timer Datasheet.
https://www.ti.com/lit/ds/symlink/na555.pdf?ts=1779184670697&ref_url=https%253A%252F%252Fwww.ti.com%252Fproduct%252FNA555

Ref5: Virtual grounds and examples.
<https://tangentsoft.com/elec/vgrounds.html>

Ref6: Potentiometers.
<https://www.jotrin.com/technology/details/potentiometer-pinout-symbol-wiring-and-working?srsltid=AfmBOorvFBTV3ZwuU9K10ig9qOE0nMZmXfOxQ9t7uRzYsiRLWB0Qjqkc>

Ref7: Inverting OPAMP topology.
<https://electrosome.com/inverting-amplifier-using-opamp/>

Ref8: RMS Formulae:
<https://www.electronics-tutorials.ws/accircuits/rms-voltage.html>

Ref9: Tim Stinchcomb Tri to Sine.
<https://www.timstinchcombe.co.uk/index.php?pge=trisin>

Ref10: TI Analog Engineer's Pocket Reference.
<https://www.ti.com/product-category/amplifiers/analog-engineers-pocket-reference-guide.html>
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